

VASHISTH TIWARI

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EDUCATION

Carnegie Mellon University

Pittsburgh, PA

Ph.D. in Language Technologies Institute, School of Computer Science

Aug 2025 –

- **Advisors:** Emma Strubell, Zico Kolter
- Research Interests: Reinforcement Learning for LLMs and improving its sample efficiency and reward attribution, better synthetic data generation, understanding RL training dynamics to inform more efficient alternatives like SFT.

Master of Science in Artificial Intelligence Engineering—ECE (GPA: 3.97/4.0)

Dec 2024

- **Courses:** Deep Learning Systems, Advanced NLP, Deep Learning, Machine Learning, Stochastic Processes

University of Rochester

Rochester, NY

Bachelor of Science in Physics, Bachelor of Arts in Mathematics (GPA: 3.97/4.0)

May 2023

- **Courses:** Probability, Hons. Real Analysis, Hons. Linear Algebra, Multi-Dimensional Calculus, Differential Equations

EXPERIENCE

Nvidia

Santa Clara, CA

Research Intern (Advisor: Oleksii and Adi)

May 2026 –

- Post Training for Nemotron

Google

Seattle, WA

Student Researcher (Advisor: Dr. Giulia DeSalvo)

May 2025 – Aug 2025

- Post Training for Gemma models, primarily focusing on reinforcement learning and improving the generation of synthetic data for supervised finetuning.
- Extended SoftSRV, a method to generate better synthetic data, to non-Gemma models, exploring cross-architecture generalization.

Carnegie Mellon University

Pittsburgh, PA

Research Staff, Improved Synthetic Data Generation (Advisor: Prof. Emma Strubell)

Feb 2025 – May 2025

- Designed a responsive-feedback driven framework where teacher models iteratively refine synthetic data generation based on student model's performance and learning outcomes (for math and reasoning tasks in post training).

Carnegie Mellon University

Pittsburgh, PA

Research Assistant, Efficient ML (Advisor: Prof. Beidi Chen)

Feb 2024 – Aug 2024

- Showcased how speculative decoding can mitigate the tradeoff between throughput and latency in LM inference.
- Implemented weight pruning, attention sparsity, and activation sparsity techniques for drafting in self-speculation.
- Evaluated compressed models' effectiveness through acceptance rates and speedup metrics in speculative decoding.
- Achieved up to a 2x speedup over auto-regressive baseline for LLaMA-3-8B inference at high batch sizes (≥ 128) through self-speculation and sparse key-value optimizations.

Mana Finance Corporation

Hillsborough, CA

Quantitative Research Intern (Mentor: Max Novendstern, David Kaufman)

May 2022 – Aug 2022

- Utilized statistical techniques to analyze stock price distributions and quantify investment risk.
- Developed ML models using Facebook Prophet for assessing expected yields on potential investments.

Los Alamos National Laboratory

Los Alamos, NM

Research Intern (Advisor: Dr. Malcolm Boshier)

Jun 2021 – Aug 2021

- Modeled complex quantum system using Python and utilized Mathematica numerical differential equation solvers.
- Discovered optimal laser pulse parameters through high-dimensional optimization.
- Improved the system performance by 5% beyond the current state-of-the-art pulse parameters through optimized pulses.

PUBLICATIONS & MANUSCRIPTS

(* denotes equal contribution.)

MACHINE LEARNING

- Eungyeup Kim*, **Vashisth Tiwari***, Chenchen Gu*, Zico Kolter. "Reliable and Sample-Efficient Evaluation of LLM Reasoning via Rare-Event Estimation." *Submitted Manuscript* (2026)
- Jared Fernandez*, Clara Na*, **Vashisth Tiwari***, Yonatan Bisk, Sasha Luccioni, Emma Strubell. "Energy Considerations of Large Language Model Inference and Efficiency Optimizations." **ACL (Main Conference) 2025**.
- Jian Chen*, **Vashisth Tiwari***, Ranajoy Sadhukhan*, Zhuoming Chen, Jinyuan Shi, and collaborators. "MagicDec: Breaking the Latency-Throughput Tradeoff for Long Context Generation." **ICLR 2025**.

PHYSICS

- Ceren Uzun, Saurabh Pandey, **Vashisth Tiwari**, Katarzyna Krzyzanowska, and Malcolm Boshier. “Improved Bragg splitting of Bose–Einstein condensates into high-order momenta wave-packets.” *APS Division of Atomic, Molecular and Optical Physics Meeting Abstracts*, 2023.
- Ceren Uzun, Saurabh Pandey, **Vashisth Tiwari**, Katarzyna Krzyzanowska, and Malcolm Boshier. “High-fidelity splitting of Bose–Einstein condensates into high-order momentum states.” *APS Division of Atomic, Molecular and Optical Physics Meeting Abstracts*, 2022.
- Amanda Wasserman, **Vashisth Tiwari**, and Segev BenZvi. “Using Machine Learning to Develop a Transient Identification Pipeline for DESI.” Presented at the *American Astronomical Society (AAS)*, 2021.

MATHEMATICS (ALPHABETICAL AUTHOR LIST)

- Anna Cusenza, Aiden Dunkelberg, Kate Huffman, Dianhui Ke, Daniel Kleber, McClatchey, Miller, Christopher Mizgerd, **Vashisth Tiwari**, Ji Ye, and Xiaoyu Zheng. “Bounds on Zeckendorf Games.” *Fibonacci Quarterly*, 60(1), 57–71, 2022.
- Anna Cusenza, Aiden Dunkelberg, Kate Huffman, Dianhui Ke, Daniel Kleber, Steven J. Miller, Christopher Mizgerd, **Vashisth Tiwari**, Ji Ye, and Xiaoyu Zheng. “Winning Strategy for Multiplayer and Multialliance Zeckendorf Games.” *Fibonacci Quarterly*, 59(4), 308–318, 2021.

TEACHING EXPERIENCE

Carnegie Mellon University	Pittsburgh, PA
<i>Teaching Assistant, Inference Algorithms for Language Modeling (11-763) w/ Prof. Graham Neubig</i>	Fall 2025
• Designed assignments and the final project on online LLM inference and accuracy-throughput tradeoff for the <i>first</i> offering of this course	
<i>Teaching Assistant, Advanced Natural Language Processing (11-711) w/ Prof. Graham Neubig</i>	Fall 2024
• Mentored research projects, held office hours, and developed a baseline RAG system for benchmarking	
University of Rochester	Rochester, NY
<i>Teaching Assistant, Quantum Theory</i>	Spring 2023
<i>Teaching Assistant, Advanced Electromagnetism</i>	Fall 2022

PROJECTS

Deep Learning Systems	Aug 2024 – Dec 2024
• Built a PyTorch-like Deep Learning library from scratch with auto differentiation, optimizer, GPU support, and advanced operators (FFT, eigendecomposition) • Implemented efficient tensor operations with CPU/GPU memory management and optimization techniques	
Consumer Hardware Pruning with Preserved Reasoning	Mar 2024 – May 2024
• Enhanced the Bonsai method (structured LLM pruning for consumer hardware) to better preserve mathematical reasoning • Developed novel task-aware pruning metrics that outperformed standard perplexity-based metrics • Demonstrated superior reasoning retention in pruned models through comprehensive evaluations	

AWARDS & HONORS

Semi-Finalist, Rhodes Scholarship , Indian Consulate	2022
Phi Beta Kappa , National Honor Society	2023
Harry W. Fulbright Prize , University of Rochester (<i>Awarded for excellence in experimental physics</i>)	2023
Undergraduate Teaching Award , Dept. of Physics & Astronomy	2023

SERVICE & LEADERSHIP

Peer Mentor , ECE Department, Carnegie Mellon University	Jan 2024 – Dec 2024
President , Society of Physics Students, University of Rochester	Aug 2021 – May 2023

SKILLS & INTERESTS

Programming	Python (Expert), C++ (Intermediate), Java (Intermediate), Bash (Proficient), Mathematica
ML/AI	PyTorch, HuggingFace, TensorFlow, LangChain, Faiss, CUDA, JAX
Data Engineering	NumPy, Pandas, Scikit-Learn, Spark, Kafka, AWS (EC2, S3, SageMaker), Docker
Interests	Competitive badminton, Tennis, Frisbee, Photography, Hiking